



Identification of tea leaf diseases by using an improved deep convolutional neural network

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ABSTRACT

Accurate and rapid identification of tea leaf diseases is beneficial to their prevention and control. This study proposes a method based on an improved deep convolutional neural network (CNN) for tea leaf disease identification. A multiscale feature extraction module is added into the improved deep CNN of a CIFAR10-quick model to improve the ability to automatically extract image features of different tea leaf diseases. The depthwise separable convolution is used to reduce the number of model parameters and accelerate the calculation of the model. Experimental results show that the average identification accuracy of the proposed method is 92.5%, which is higher than that of traditional machine learning methods and classical deep learning methods. The number of parameters and the convergence iteration times of the improved model are significantly lower than those of VGG16 and AlexNet deep learning network models.

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1. Introduction

Tea is an important economic crop. In 2017, China's tea plantations accounted for 2.849 million hectares and produced 2.46 million tons of tea leaves, which were worth approximately \$30 billion. However, tea plants are frequently infected with diseases during growth. More than 100 common diseases affect tea leaves. Fig. 1 shows the tea leaves and tea plants infected with tea leaf blight in the Tianjingshan Tea Garden, Anhui Province, China. These tea leaf diseases can cause the poor growth of tea plants, resulting in a decrease in the yield and quality of tea leaves. The annual tea leaf yield is reduced by approximately 20% due to diseases, causing serious economic losses to tea farmers. Therefore, tea leaf diseases should be accurately identified, and appropriate preventive measures should be promptly implemented to reduce tea yield loss and improve tea quality.

Traditional crop disease identification mainly relies on artificial methods, which are limited by a long cycle and strong subjectivity. With computer technology development, image processing and machine learning methods have been widely used for crop disease identification [1,2]. Sun et al. proposed an algorithm combining simple linear iterative clustering (SLIC) with a support vector machine (SVM) to extract a significant map of tea leaf diseases from

the complex background of the images of tea tree diseases. The proposed method provided a basis for further investigating tea leaf disease identification [3]. Hossain et al. analyzed 11 features of such images and used an SVM classifier to identify diseases [4]. Karmokar et al. proposed a tea leaf disease identifier to extract the features of tea leaf images and used neural network integration to identify diseases [5]. In addition to tea leaf diseases, four common leaf diseases, such as leaf spot, rust, and tail plaque, have been identified using an SVM and an image identification based pattern identification algorithm [6]. Pantazi et al. used a local binary pattern feature extraction and a class of classification methods to automatically identify leaf sample images from different crop varieties [7]. Kumar et al. presented a sugarcane crop monitoring system model that continuously monitors the temperature, humidity, and moisture of crops and used KNN clustering and SVM classifiers to identify diseases on regularly acquired images [8]. However, these traditional machine learning methods must preextract disease features in identifying crop diseases. Artificially extracted features may not reflect the essential attributes of tea leaf diseases because of the complex texture and spectral features of infected tea leaves. The accuracy of the above methods in identifying tea leaf diseases is low.

Deep learning methods do not require manual feature extraction, so they have been widely used in target identification and other fields [9–11]. These methods have also been utilized to identify crop diseases. Sun et al. applied a CNN to identify the images of tea leaf diseases. With this method, tea leaf images are segmented

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Fig. 1. Tea leaves and tea plants infected with tea leaf blight.

and enhanced, and they are used to train CNN. The identification accuracy of tea leaf diseases is improved by adjusting the network parameters [12]. Guan et al. compared the classification effects of VGGNet, Inception-v3, and ResNet50 networks and the performance of shallow networks trained from scratch and deep models finetuned through transfer learning on the severity of apple black rot [13]. Zhang et al. proposed two models, namely, GoogLeNet and Cifar10 model, for leaf disease identification. These models are used to train and test nine kinds of maize leaf images [14]. Mohanty et al. utilized AlexNet and GoogLeNet networks to identify 26 diseases of 14 crops [15]. Ramcharan et al. applied transfer learning to train a deep CNN for identifying three diseases and two pests of cassava [16]. Although the performance of these CNNs in identifying crop diseases is good, these models are limited by various disadvantages, particularly the requirement of many parameters and a slow calculation speed.

A CIFAR10-quick model is a deep learning model derived from MatConvNet with few parameters and low computational costs [17]. However, this model cannot effectively extract image features, and its identification accuracy is insufficient. In this study, the CIFAR10-quick model is improved. Convolution kernels with different sizes are constructed in convolution layers, which are used to extract the multiscale features of the images of tea leaf diseases. Standard convolution is replaced by depthwise separable convolutions, which are used to reduce the number of model parameters and improve the calculation speed of the model. The proposed identification method can rapidly and accurately identify tea leaf diseases.

2. Materials and methods

2.1. Data acquisition

The images of the tea leaf were captured with a Canon EOS 80D SLR camera, and the selected location was the Tianjingshan Tea Garden, which is located in Anhui Province, China, at 31.14°37'N and 117.36°16'E and 40 m above sea level.

Fig. 2 shows healthy tea leaves and three kinds of infected tea leaves. The tea leaves are infected with tea leaf blight, tea bud blight, and tea red scab. A total of 36 images are found in each category. Training a deep CNN model requires a large amount of data samples, but the number of the collected original images of the tea leaf diseases is insufficient. Training samples should be augmented to increase the number of samples in the original data set. After a series of preprocessing, such as denoising the original images of the tea leaf diseases, 26 images are randomly selected from each category of the images for 90°, 180°, and 270° rotations, up-and-down swapping, and left-right swapping for data augmentation. The augmented images are used as training samples for a CNN model. The 10 remaining images of each category are used to test the identification accuracy of the model after training.

Table 1
Specific parameters of the CIFAR10-quick model used in this study.

Layer	Kernel or filter size, output channel	Output size
Convolution	5*5,32	32*32*32
Max pooling	2*2,32	16*16*32
Convolution	5*5,32	16*16*32
Max pooling	2*2,32	8*8*32
Convolution	5*5,32	8*8*32
Max pooling	2*2,32	4*4*32
Fully connected	-,512*64	64
Fully connected, SoftMax	-,64*4	4

2.2. Deep CNN model

2.2.1. CIFAR10-quick model

The CIFAR10-quick model consists of convolution, maximum pooling, fully connected, and SoftMax layers. The input of the model is an RGB image with a size of 32*32. Convolution layers use convolution kernels with a side length of 5. Maximum pooling layers involve filters with a side length of 2. The CIFAR10-quick model has a simple structure and a fast calculation ability, which satisfy the requirements for the rapid identification of tea leaf diseases. Since this study needs to identify healthy tea leaves and three kinds of infected tea leaves, the adapted CIFAR10-quick model has four neurons at output layer. The structure and specific parameters of the CIFAR10-quick model used in this study are shown in Fig. 3 and Table 1, respectively.

2.2.2. Multiscale feature extraction

The convolution kernels of the CIFAR10-quick model have a size of 5*5, and the extracted features are insufficiently distinguishable from different diseases. In this study, 5*5 convolution kernels in the CIFAR10-quick model are replaced by 3*3 and 7*7 convolution kernels for the multiscale feature extraction of images of tea leaf diseases. The schematic of multiscale feature extraction is shown in Fig. 4.

Convolution kernels with different sizes have various sizes of receptive fields, which can acquire features with different scales. Replacing the original convolution kernel with large and small convolution kernels can help the model to extract multifield features of images of tea leaf diseases. In Fig. 4, the input layer uses convolution kernels with side lengths of 3 and 7 for forward propagation. Although the convolution kernels have different sizes, they are filled with 0, and the step size is set to 1. Thus, the length and width of the output matrix obtained through forward propagation are the same. The output results of convolution kernels with different sizes are connected in depth to achieve the multiscale fusion of features. The tea leaves infected with different diseases slightly vary. In addition, the backgrounds of the images of the infected tea leaves are complex. Thus, the multiscale feature extraction of the images of the infected tea leaves can improve the distinguishability of the model for different tea leaf diseases.

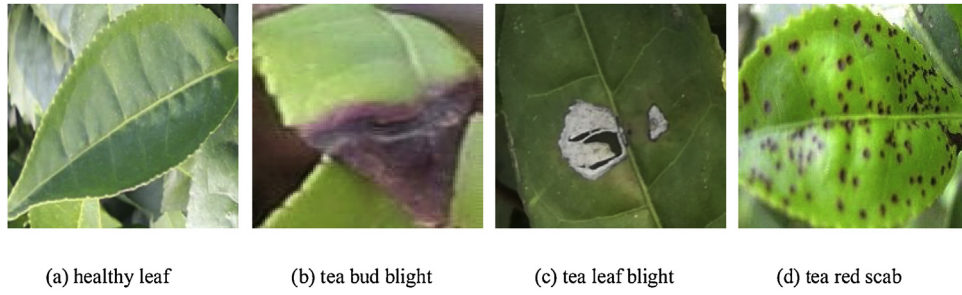


Fig. 2. Healthy and infected tea leaves.

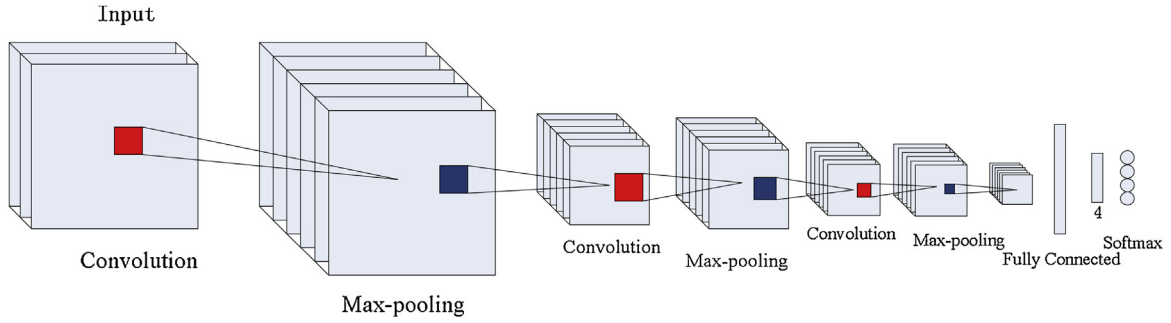


Fig. 3. Structure of the CIFAR10-quick model used in this study.

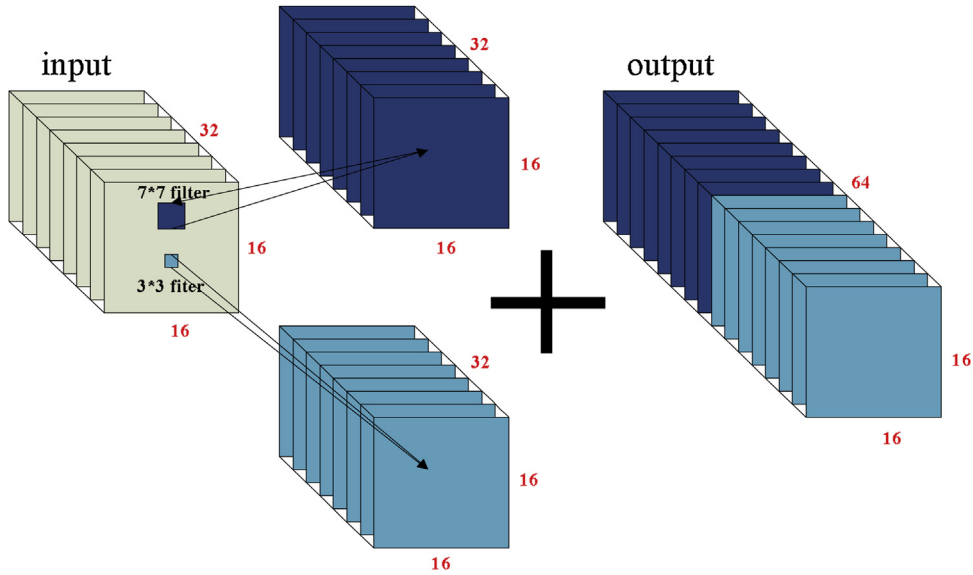


Fig. 4. Schematic of multiscale feature extraction.

2.2.3. Depthwise separable convolution

Depthwise separable convolution is used to replace the standard convolution. Depthwise separable convolution is obtained from MobileNet [18,19], which not only exhibits less parameters and lower computational complexity than the standard convolution but also improves model performance to some extent.

Depthwise separable convolution decomposes standard convolution into depthwise and pointwise convolutions. In Fig. 5, the M channels of the input for the input feature map with a size of (D, D, M) are initially filtered using the M convolution kernels of size $(K, K, 1)$. The filtered results are combined to complete the depthwise convolution. Then, the results of depthwise convolution are convolved using a convolution kernel with a size of $(1, 1, M)$ to perform a pointwise convolution. Depthwise separable convolution is equivalent to learning channel and spatial features separately.

The number of parameters in a convolutional layer is $M * K * K * d + b_k \approx M * K * K * d$, where b_k is the number of parameters of bias and d is the depth of convolution kernels. For standard convolutions, the depth of convolution kernels is the same as the number of convolution kernels, namely $d = M$, so the number of parameters in a standard convolutional layer is $M^2 * K^2$.

Depthwise separable convolution decomposes standard convolution into depthwise and pointwise convolutions. For depthwise convolutions, the depth of convolution kernels is $d = 1$. For pointwise convolutions, the width and length of convolution kernels are both 1, that is, $K = 1$. Then the number of parameters in a depthwise separable convolutional layer is $M * K * K * 1 + M * 1 * 1 * d = M * K^2 + M^2$. The number of parameters of depthwise separable convolution is lower than that of standard convolution. Therefore, depthwise separable convolution can solve the problem that the

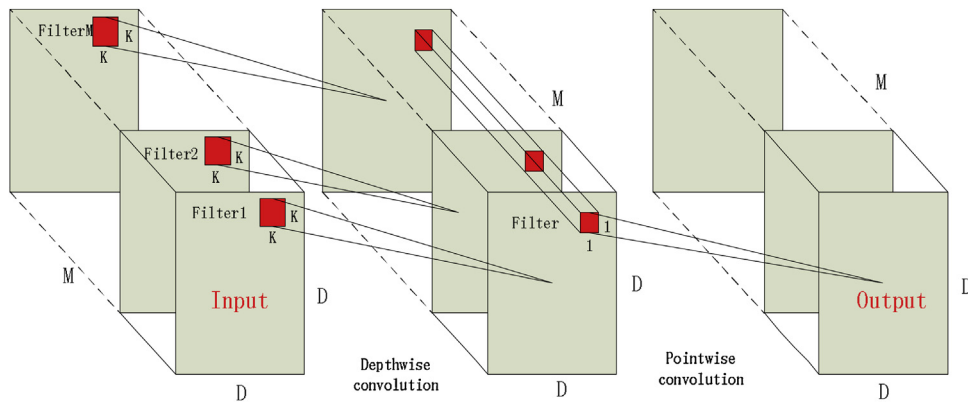


Fig. 5. Schematic of depthwise separable convolution.

Table 2

Identification accuracy and loss value of four models with different structures after 5000 iterations.

Model	Alter-first	Alter-second	Alter-third	Alter-all
Identification accuracy	92.5%	92.5%	70%	82.5%
Loss value	0.35	0.002	0.17	0.001

number of parameters increases because of the introduction of the multiscale feature extraction module.

2.2.4. Model training

In this study, a *relu* activation function is used for the convolutional and fully connected layers of the model. This function can adaptively learn the parameters of a rectifier. The output of CNN is converted into a probability distribution by using the SoftMax function. An average cross entropy loss function is used to measure the difference between the prediction result and the label of input samples. An Adam algorithm is used to optimize the loss function.

2.2.5. Model structure selection

Models with different structures have different abilities to distinguish the features of the infected tea leaves. Four models with different structures can be obtained by replacing the first, second, third, and all three convolutional layers of the CIFAR10-quick model with the multiscale feature extraction module. In this study, the four models are named as Alter-first, Alter-second, Alter-third, and Alter-all, respectively. The most effective model is selected through experiments. The procedure of selection is as follows:

- Step 1: Training samples and test samples are resized to 32*32;
- Step 2: The learning rate is set to 0.0001, the batch size is set to 16, and the number of iterations is 5000;
- Step 3: The models are trained and loss value is recorded every 2 training iterations;
- Step 4: The models are tested and identification accuracy for test samples is recorded every epoch.
- Step 5: The model with highest identification accuracy and lowest loss value is selected as the most effective model.

In the experiments, the average cross entropy loss function and the Adam optimization algorithm are used. Fig. 6 shows the training loss curves of the four models and the accuracy curves of the test set. Table 2 shows the identification accuracy of the test set and the loss values of the four models after 5000 iterations.

The results show that the Alter-second model can effectively extract the features of different infected tea leaves and recognize tea leaf diseases based on the extracted features. The average identification accuracy of the Alter-second model is 92.5%. Therefore, the Alter-second model is selected as the best model in this study. The structure of the Alter-second model is shown in Fig. 7.

3. Experimental results and analysis

3.1. Comparison of the improved model and the original model

The improved model in Fig. 7 is compared with the original CIFAR10-quick model [17], and the results are shown in Fig. 8. With the same number of iteration times, the average identification accuracies of the improved model and the CIFAR10-quick model are 92.5% and 67.5%, respectively. The final loss values of the training of the improved model and the CIFAR10-quick model are 0.002 and 0.84, respectively. The performance of the original CIFAR10-quick model is lower than that of the improved model. Fig. 9 shows two tea leaf disease images identified correctly by the improved model and incorrectly by the original CIFAR10-quick model.

In Fig. 9, the area of the disease spots on the disease images is small, and the color and texture of the disease spots are similar to the background, or the disease spots are curled. Figs. 10 and 11 are the feature maps of two tea leaf disease images in Fig. 9 which are extracted by singlescale or multiscale feature extraction and then superimposed. The original CIFAR10-quick model has a weak feature extraction ability and cannot identify these disease images correctly. The improved model introduces a multiscale feature extraction module, which can extract the features of disease leaves from different scales and enhance the discriminability between different tea leaf diseases. The improved model can identify these disease images successfully.

3.2. Comparison of the proposed method and the traditional machine learning methods

Fig. 12 shows the comparison of the average identification accuracies of BP neural network [1], Bayesian classifier [2], SVM [4], KNN [8], and the proposed method. The experimental results show that the identification accuracy of the traditional machine learning methods is lower than that of the proposed method.

3.3. Comparison of the improved model and the classical CNN models

The classical CNN models used in the experiments include LeNet-5 [9], AlexNet [15], and VGG16 [13]. The average accuracy curves on the test set of the improved model and the classical CNN models are shown in Fig. 13. After 5000 iterations, the average identification accuracies of the improved model, LeNet-5, AlexNet, and VGG16 are 92.5%, 57.5%, 70%, and 87.5%, respectively. The average identification accuracy of the improved model is higher than that of LeNet-5, AlexNet, and VGG16 CNN models.

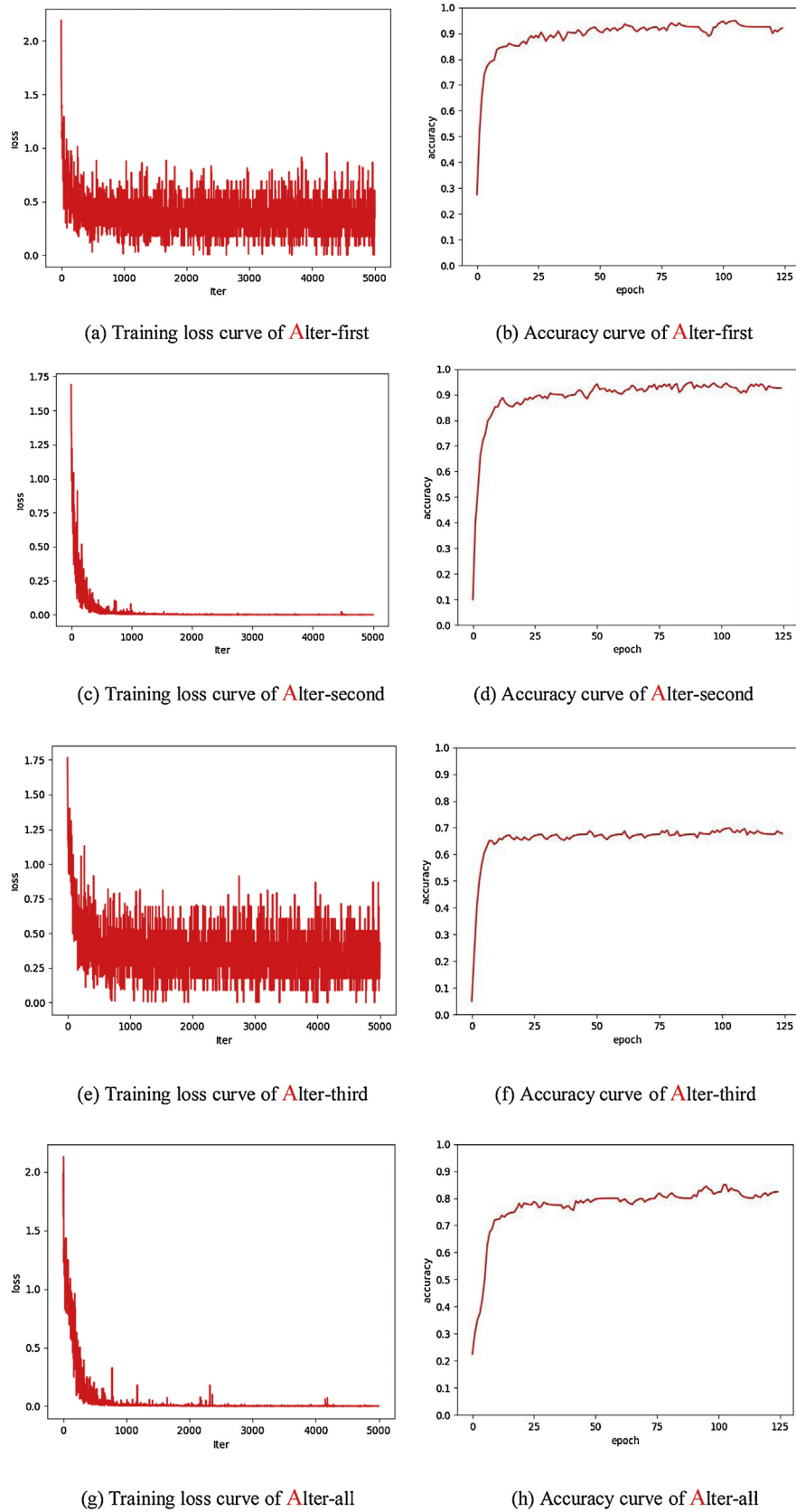


Fig. 6. Training and testing results of four models with different structures.

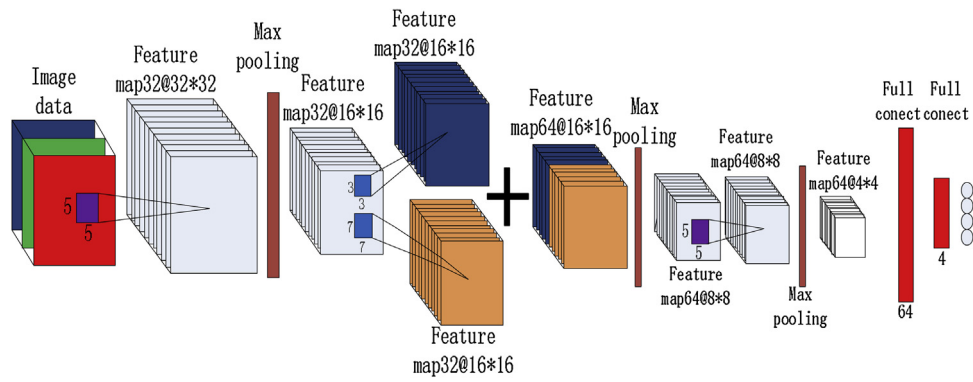


Fig. 7. Structure of the Alter-second model (note: represents the standard convolution kernel, refers to depthwise separable convolution kernels, and denotes SoftMax classifiers).

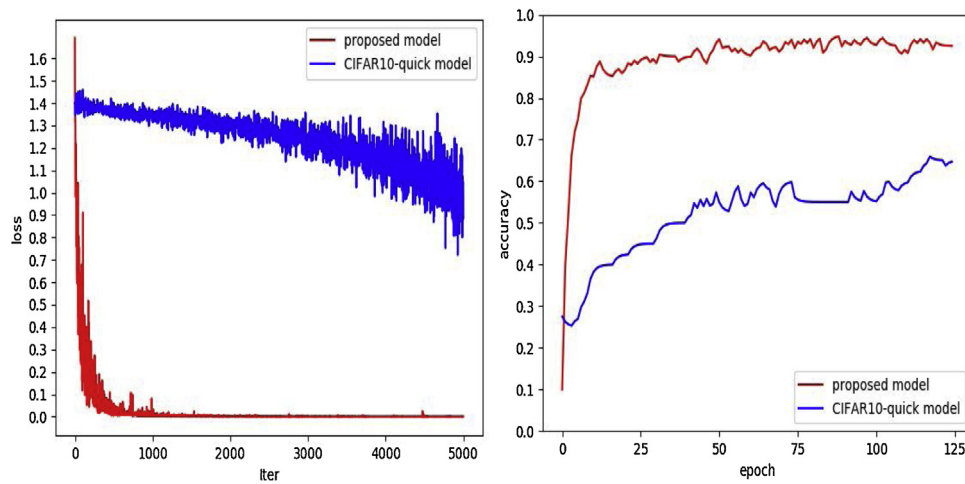


Fig. 8. Comparison of the improved model and original CIFAR10-quick model. Left is training loss curves, right is accuracy curves.



Fig. 9. Two tea leaf disease images identified correctly by the improved model and incorrectly by the original CIFAR10-quick model.

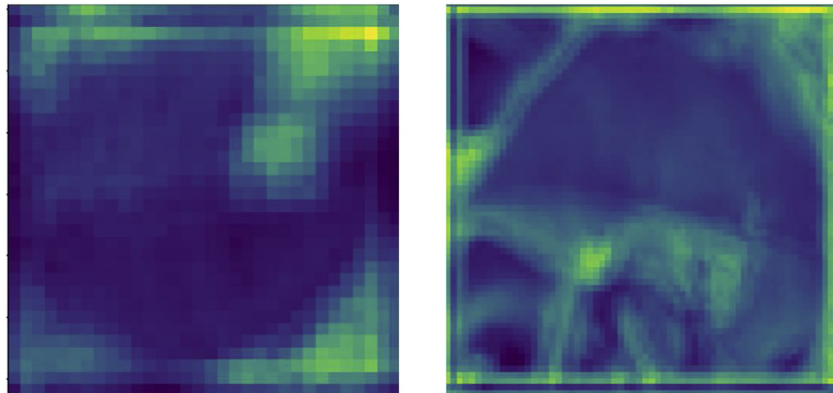


Fig. 10. Feature maps of the two tea leaf disease images in Fig. 9 by using singlescale feature extraction.

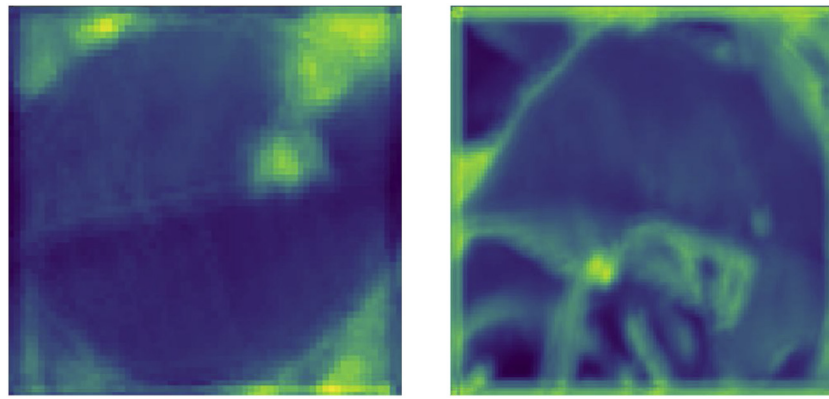


Fig. 11. Feature maps of the two tea leaf disease images in Fig.9 by using multiscale feature extraction.

Table 3

Number of parameters of different CNN models.

Model	Proposed	LeNet-5	AlexNet	VGG16
Number of parameters (10,000)	Approximately 20	Approximately 6	Approximately 6,000	Approximately 13,800

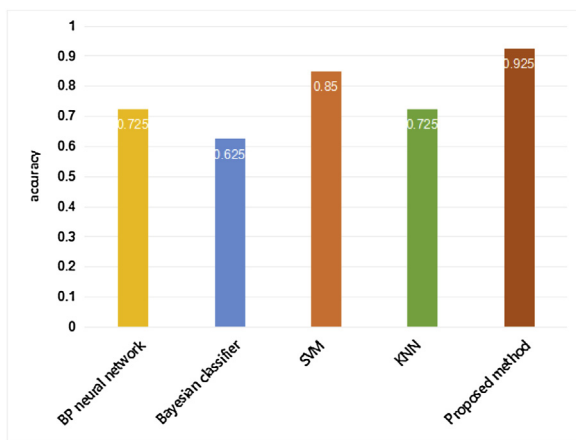


Fig. 12. Average identification accuracy of the proposed method and the traditional machine learning methods.

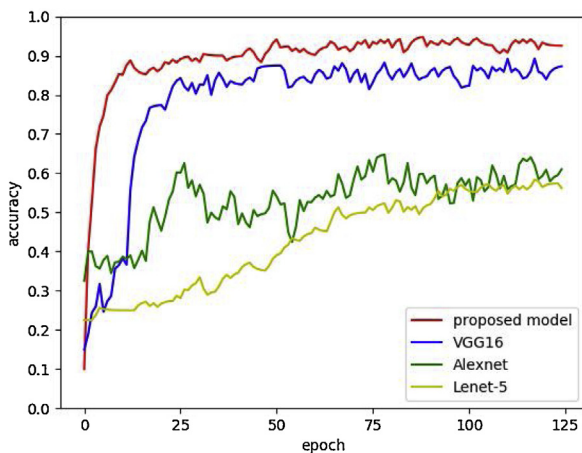


Fig. 13. Average accuracy curves of the improved model and the classical CNN models.

Table 3 shows the parameters of the improved model and the three classical CNN models. In Fig. 13 and Table 3, in comparison with the traditional CNN models, the improved model has the advantages of few parameters and high identification accuracy.

4. Conclusion

Traditional machine learning methods for identifying plant diseases require the manual extraction of the features of disease images. An advantage of deep learning in plant disease identification is that it can automatically extract the essential features of disease images. In this study, a multiscale feature extraction module is added to the CIFAR10-quick deep learning model to improve the ability to automatically extract image features of different tea leaf diseases. The standard convolution in the multiscale feature extraction module is changed to depthwise separable convolution to reduce the number of model parameters and hasten the calculation of the model. The experimental results show that the proposed model has advantages of few parameters, high identification accuracy, and fast identification speed. The average identification accuracy of the proposed model for healthy tea, tea bud blight, tea leaf blight, and tea red scab is 92.5%, which is higher than that of traditional machine learning methods, such as BP neural network, Bayesian classifier, SVM, and KNN, and classical CNN methods, such as LeNet-5, AlexNet, and VGG16.

Declaration of Competing Interest

The authors declared that they have no conflicts of interest to this work. We declare that we do not have any commercial or associative interest that represents a conflict of interest in connection with the work submitted.

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